

Md Zabirul Islam

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PROFESSIONAL SUMMARY

Ph.D. researcher in Computer Science at Rensselaer Polytechnic Institute developing **structured, verifiable, and trustworthy AI** for consequential knowledge workflows. Research spans **large language models (LLMs)**, **vision-language models (VLMs)**, evidence-grounded generation, constrained decoding, retrieval, abstention and verification, multimodal medical AI, and instructor-governed AI for education. Work includes cross-institutional clinical AI, patient-facing safety evaluation, benchmark design, and production deployment on institution-operated GPU infrastructure.

Google Scholar reports **1,254 citations**, **h-index 7**, and **i10-index 6**. Experienced with **PyTorch**, **Hugging Face**, **vLLM**, **LoRA/SFT**, **RAG**, **schema-constrained generation**, **LLM-as-judge evaluation**, and **H100-scale inference**. F-1 CPT/OPT eligible.

EDUCATION

Rensselaer Polytechnic Institute (RPI) <i>Ph.D. in Computer Science, Advisor: Prof. Ge Wang</i> – Research emphasis: LLMs, multimodal foundation models, trustworthy AI, medical imaging and clinical AI, and AI-enabled education.	Troy, NY Jan. 2024 – Present
Rensselaer Polytechnic Institute (RPI) <i>M.S. in Computer Science</i>	Troy, NY Dec. 2025
Khulna University of Engineering and Technology (KUET) <i>B.Sc. in Computer Science and Engineering</i>	Khulna, Bangladesh Mar. 2020

RESEARCH AND TEACHING EXPERIENCE

Graduate Research Assistant <i>Wang-AXIS Lab, Rensselaer Polytechnic Institute</i> – Conduct Ph.D. research on trustworthy generative and multimodal AI for healthcare and education, with emphasis on structured outputs, evidence control, verification, auditability, and human governance. – Collaborate across RPI and clinical research teams at the NIH Clinical Center , Yale School of Medicine , Massachusetts General Hospital/Harvard Medical School , and Wake Forest University School of Medicine on patient-facing AI safety and structured radiology reporting. – Develop and evaluate local/open-weight LLM and VLM systems using H100 GPUs , PyTorch , Hugging Face , vLLM , LLaMA-Factory , LoRA/SFT , RAG , NLI verification , constrained decoding , and structured evaluation pipelines . – Build production research systems that combine retrieval, multimodal generation, speech, learning analytics, and governance controls while keeping sensitive student or clinical data on institution-operated infrastructure.	Jun. 2025 – Present Troy, NY
Graduate Teaching Assistant <i>Rensselaer Polytechnic Institute</i> – Supported <i>CSCI 6360 – Parallel Computing and Parallel Programming</i> and <i>CSCI 2300 – Introduction to Algorithms</i> ; guided students in CUDA , MPI , parallel algorithm design , profiling , and performance optimization on the AiMOS supercomputer. – Supported office hours, grading, and solution development while reinforcing asymptotic analysis, dynamic programming, graph algorithms, proofs of correctness, and complexity analysis.	Jan. 2024 – May 2025 Troy, NY
Lecturer <i>Daffodil International University</i> – Taught undergraduate courses in Artificial Intelligence , Digital Image Processing , Algorithm Analysis and Design , and Numerical Methods to 1,000+ students. – Mentored 30+ students in research-oriented projects and supervised technical coursework spanning machine learning, algorithms, and image analysis.	Feb. 2021 – Dec. 2023 Dhaka, Bangladesh

- MIP-BENCH: Instructional Grounding Benchmark for Medical Imaging VLMs** Jan 2026 – Present
- *Manuscript in preparation for submission to ICML.*
 - Introduced a benchmark for whether VLMs recover **instructional intent**, not merely visual or semantic correctness, from **1,117 slides across 23 medical imaging lectures and 15 scientific domains**.
 - Proposed the **Instructional Grounding Score (IGS)** combining Concept Recall, Relational Validity, and Scope Fidelity; validated the metric against graph-based structural analysis, a second LLM judge, and pooled human ratings.
 - Evaluated five open-weight VLMs from 7B–40B parameters; the strongest model reached **IGS = 0.642** versus pooled human **IGS = 0.690**, while all five showed a consistent **scope–concept gap of +0.10 to +0.14**.
 - Built a curriculum knowledge graph with **3,712 concept nodes and 12,419 co-occurrence edges**; released benchmark splits, annotations, model outputs, and evaluation protocols on GitHub and Hugging Face.
- Fully Structured Lung Nodule Reporting with Large Language Models** Sep. 2024 – Present
- *Co-first author; manuscript submitted to the Journal of the American Medical Informatics Association (JAMIA).*
 - Developed and cross-institutionally validated **dynamic template-constrained decoding** for converting free-text low-dose CT lung cancer screening reports into fully structured lung nodule reports.
 - Evaluated **5,442 reports from two institutions** with two manually structured 250-report cohorts; achieved **F1 = 0.976 and 0.969** with no formatting errors or schema violations in the primary validation.
 - Fine-tuned six model families on approximately **30,000 synthetic reports** built from real report language; compact models approached the strongest 70B prompt-only model while reducing token expenditure per report by **84.2–87.2%**.
 - Added schema-guided grammar constraints that increased raw JSON-schema validity from **97.7% to 100%** across 1,500 model-report outputs with less than one percentage point change in micro-F1.
- Structural Safety of AI for Patient Onboarding in Clinical Trials** Mar 2026 – Present
- *Manuscript in preparation for submission to npj Digital Medicine.*
 - Designed an auditable **trial-first single-context pipeline** that selects one dominant clinical trial before generation or abstains, then restricts eligibility triage, consent-style explanation, and teach-back to evidence from that trial.
 - Evaluated **114 onboarding cases across six model families**: two lexical detectors found **0/684 cross-trial leaks**; dual-LLM semantic consensus flagged 12/684 (1.75%), with manual review identifying eight genuine paraphrased cross-trial errors.
 - Demonstrated that standard multi-trial RAG, prompt-only guarding, and citation enforcement produced **50.9–59.0% semantic-consensus leakage**, whereas structural single-trial systems remained at **1.3–1.8%**.
 - Implemented claim-level **DeBERTa NLI verification** that removed all eight manually confirmed semantic leaks at a **15.1% per-claim utility cost**; positioned the system for coordinator-supervised preclinical onboarding rather than autonomous eligibility decisions.
- ALIVE: Instructor-Governed AI for Closed-Loop Education** Jun. 2026 – Present
- *Manuscript in preparation for submission to Nature Communications.*
 - Built and deployed a **closed-loop AI education system** in which an instructor-governed AI delivers lectures, asks formative questions, evaluates spoken/free-text answers, provides feedback, transfers demonstrated weaknesses to a one-to-one tutor, and tracks recurring misconceptions for course revision.
 - Deployed across **two biomedical imaging courses, 48 lectures, and 2,067 slides**; the classroom and tutor share a course-adapted **31B multimodal model**, cloned-voice TTS, and retrieval service on institution-operated hardware.
 - Constructed graph-structured retrieval over a **102-document course corpus**; relevant context improved gold-token recall by **+0.241** over instruction-matched irrelevant context. LoRA adaptation on **37,236 gated course-derived pairs** improved held-out recall by **+0.056 and +0.102** on two evaluation sets.
 - Implemented approval gates, audit trails, per-student mastery state, six question types, deterministic misconception tracking, instructor overrides, and an explicit data-use switch so consequential course changes remain under human control.

Submitted / in preparation

- **Md Zabirul Islam**, *et al.*, “MIP-BENCH: A Benchmark for Instructional Grounding in Medical Imaging Presentations.” *Submitted to NeurIPS 2026 Evaluations and Datasets Track*, 2026.
- Md Sayed Tanveer*, **Md Zabirul Islam***, Chuang Niu, *et al.*, “Development and Validation of a Large Language Model for Generating Fully-Structured Lung Nodule Reports.” *Submitted to JAMIA*, 2026. *Equal contribution.
- **Md Zabirul Islam**, Darko Pucar, Christopher Whitlow, Chi Liu, Ge Wang, “Structural Safety of AI for Patient Onboarding in Clinical Trials Using a Grounded Pipeline.” *Manuscript in preparation for npj Digital Medicine*, 2026.
- **Md Zabirul Islam**, Ge Wang, “Closing the Instructional Feedback Loop with Instructor-Governed AI.” *Manuscript in preparation for Nature Communications*, 2026.

Peer-reviewed / published

- Md Motaleb Hossen Manik, **Md Zabirul Islam**, Ge Wang, “MILU: A Consensus Ensemble Benchmark for Multimodal Medical Imaging Lecture Understanding,” *Journal of Medical Imaging*, vol. 13, no. 6, 062202, 2026.
- Md Motaleb Hossen Manik, **Md Zabirul Islam**, Ge Wang, “MEDI-SLATE: Medical Imaging Slide-Lecture Aligned Teaching Ensemble,” *Visual Computing for Industry, Biomedicine, and Art*, vol. 9, article 8, 2026.
- Md Motaleb Hossen Manik, William Muldowney, **Md Zabirul Islam**, Ge Wang, “Development of an Optically Emulated Computed Tomography Scanner for College Education,” *Visual Computing for Industry, Biomedicine, and Art*, vol. 9, article 2, 2026.
- **Md Zabirul Islam**, Ge Wang, “Avatars in the Educational Metaverse,” *Visual Computing for Industry, Biomedicine, and Art*, vol. 8, article 15, 2025.
- Ling Yue, Sixue Xing, Jonathan Li, **Md Zabirul Islam**, Bolun Xia, Jintai Chen, Tianfan Fu, “TrialDura: Hierarchical Attention Transformer for Interpretable Clinical Trial Duration Prediction,” *NeurIPS 2024 Workshop on AI for New Drug Modalities*, 2024.
- **Md Zabirul Islam**, Md Milon Islam, Amanullah Asraf, “A Combined Deep CNN-LSTM Network for COVID-19 Detection Using X-ray Images,” *Informatics in Medicine Unlocked*, vol. 20, 100412, 2020.
- Md Milon Islam, **Md Zabirul Islam**, Amanullah Asraf, *et al.*, “Diagnosis of COVID-19 from X-rays Using Combined CNN-RNN Architecture with Transfer Learning,” *BenchCouncil Transactions on Benchmarks, Standards and Evaluations*, vol. 2, no. 4, 100088, 2022.
- Amanullah Asraf, **Md Zabirul Islam**, Md Rezwanul Haque, Md Milon Islam, “Deep Learning Applications to Combat Novel Coronavirus (COVID-19) Pandemic,” *SN Computer Science*, vol. 1, article 363, 2020.

Full publication list: Google Scholar

TECHNICAL SKILLS

Generative and Multimodal AI: LLMs, VLMs, RAG, LoRA/SFT, constrained decoding, schema-guided generation, structured outputs, LLM-as-judge, NLI verification, knowledge graphs, prompt engineering, multimodal fusion.

Models and Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, vLLM, LLaMA-Factory, TorchTune, LangChain, FAISS, BGE-M3; Llama, Qwen, Mistral, Gemma, InternVL, LLaVA, Whisper.

Evaluation: F1/micro-F1, BERTScore, ROUGE, BLEU, NDCG, bootstrap confidence intervals, inter-rater agreement, ablation studies, benchmark design, structured safety evaluation.

Medical and Clinical AI: CT/LDCT, radiology reporting, DICOM workflows, structured clinical text, Lung-RADS-oriented reporting, clinical trial retrieval and evidence grounding.

Systems and Deployment: CUDA, NVIDIA H100, AiMOS, DGX Spark, MPI, vLLM serving, FastAPI, REST APIs, Docker, Linux/Ubuntu, model optimization, multi-GPU inference.

Programming: Python, C, C++, Java, JavaScript, Bash, SQL, R, MATLAB.